

Exact Two-Transaction Sampling Policies for Risk-Limiting Financial Audits

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Abstract

Shekhar, Xu, Lipton, Liang, and Ramdas derive a one-step oracle rule for risk-limiting financial audits: sample a remaining transaction i with probability proportional to its monetary misstatement contribution $\pi_i f_i$. They do not claim that repeating this rule minimizes total expected audit length. For their released ApproxKelly initialization intersected with the logical confidence sequence, we solve the global sampling problem exactly when $N = 2$. For every first-stage distribution q ,

$$\mathbb{E}_q[\tau] = 2 - \sum_{i: 1-\pi_i \leq \varepsilon} q(i).$$

Thus the optimum is 1 if $\varepsilon \geq \min\{\pi_1, \pi_2\}$, and 2 otherwise. An exact rational instance has expected stopping times 1, $5/4$, and $3/2$ under the deterministic optimum, prop-M, and the repeated oracle, respectively. We also characterize strict-support conventions and give an infinite rational family.

1 Setup and the one-step oracle

Consider N transactions with normalized recorded-value weights $\pi_i > 0$, where $\sum_i \pi_i = 1$, and fixed misstatement fractions $f_i \in [0, 1]$. The total monetary misstatement is

$$m^* = \sum_{i=1}^N \pi_i f_i.$$

At time t , an auditor samples one remaining transaction from a predictable distribution q_t , observes its f_i , updates a confidence sequence, and stops when the confidence-set diameter is at most ε .

Shekhar et al. [1] prove that, conditional on the current history, the importance-weighted next observation has minimum variance under

$$q_t^{\text{oracle}}(i) = \frac{\pi_i f_i}{\sum_{j \in R_t} \pi_j f_j}.$$

This rule also maximizes a lower bound on one-step test-wealth growth. The paper describes the full fastest-uncertainty-reduction problem as a difficult multistage optimization and solves the one-step surrogate instead. We study the natural derived question of whether repeating this oracle minimizes $\mathbb{E}[\tau]$. This question is not stated as a conjecture in [1].

2 The pinned confidence construction

For a candidate total m , the paper's betting process has update

$$W_t(m) = W_{t-1}(m) \{1 + \lambda_t(m)(Z_t - \mu_t(m))\}, \quad W_0(m) = 1,$$

and retains candidates with $W_t(m) < 1/\delta$. The logical interval after history $I_1, \dots, I_t$ is

$$[L_t, U_t] = \left[\sum_{s=1}^t \pi_{I_s} f_{I_s}, \sum_{s=1}^t \pi_{I_s} f_{I_s} + \sum_{i \in R_{t+1}} \pi_i \right].$$

We pin the construction to the authors' released implementation [2]: the ApproxKelly previous-payoff accumulator is initialized at zero, its first bet is $\lambda_1 = 0$, and the betting bounds are intersected with the logical interval and their running intersection. The logical refinement is equation (6) of [1].

Lemma 1 (First-round reduction). *For every $\delta \in (0, 1)$, the first-round betting confidence set in the pinned construction is $[0, 1]$. After sampling $I_1 = i$, the combined confidence-set diameter is $1 - \pi_i$.*

Proof. Since $\lambda_1 = 0$, one has $W_1(m) = 1$ for every $m \in [0, 1]$. The inequality $\delta < 1$ gives $1 < 1/\delta$, so the betting process rejects no candidate. Intersecting with the logical interval leaves an interval whose diameter is the total remaining weight, namely $1 - \pi_i$. $\square$

3 Admissible actions and support

Definition 2 of [1] defines q_t as a probability distribution on the remaining set. Proposition 2 optimizes over the simplex of such distributions, and Remark 5 discusses deterministic sampling. The literal action space therefore includes boundary distributions.

There is nevertheless a support subtlety. The generic identity

$$\mathbb{E} \left[\frac{\pi_{I_t} f_{I_t}}{q_t(I_t)} \middle| \mathcal{F}_{t-1} \right] = \sum_{i \in R_t} \pi_i f_i$$

requires positive mass on each item with positive contribution. We therefore state results both for the literal simplex and for conservative strict-support conventions. In the deterministic $N = 2$ optimum below, the first bet is zero and the audit stops from the logical interval alone, so no later importance-weighted bet is used.

4 Exact solution for $N = 2$

For positive $\pi_1 + \pi_2 = 1$, define the first-round stopping set

$$A(\pi, \varepsilon) := \{i \in \{1, 2\} : 1 - \pi_i \leq \varepsilon\}.$$

Theorem 1 (Complete $N = 2$ characterization). *Fix any $f_1, f_2 \in [0, 1]$, $\varepsilon, \delta \in (0, 1)$, and the pinned construction. For every first-stage distribution $q \in \Delta_{\{1, 2\}}$,*

$$\mathbb{E}_q[\tau] = 2 - \sum_{i \in A(\pi, \varepsilon)} q(i). \quad (1)$$

Consequently,

$$V^* := \min_{q \in \Delta_{\{1, 2\}}} \mathbb{E}_q[\tau] = \begin{cases} 2, & \varepsilon < \min\{\pi_1, \pi_2\}, \\ 1, & \varepsilon \geq \min\{\pi_1, \pi_2\}. \end{cases} \quad (2)$$

If A is empty, every q is optimal. If A is nonempty, the minimizers are exactly the distributions satisfying $q(A) = 1$.

Proof. By the lemma, sampling item i first stops the audit exactly when $i \in A$. This event has probability $q(A)$. On its complement, the only remaining item is sampled at time two, when the logical interval is a singleton. Hence

$$\mathbb{E}_q[\tau] = q(A) + 2\{1 - q(A)\} = 2 - q(A),$$

which proves (1). A distribution can make $q(A) = 1$ exactly when A is nonempty. Finally, $A \neq \emptyset$ if and only if $\varepsilon \geq \min\{\pi_1, \pi_2\}$. This gives (2) and the optimizer set. $\square$

The optimal policy is independent of both f and δ . In the one-stopping-item regime, it audits the larger-weight transaction deterministically.

Corollary 1 (Oracle-optimality criterion). *Assume $m^* > 0$, so the proportional oracle is defined. If A is empty or $A = \{1, 2\}$, the oracle is globally optimal because every policy is optimal. If $A = \{k\}$ and $j \neq k$, then*

$$\mathbb{E}[\tau(q^{\text{oracle}})] = 2 - \frac{\pi_k f_k}{\pi_1 f_1 + \pi_2 f_2}.$$

It equals the literal-simplex optimum 1 exactly when $\pi_j f_j = 0$. If both contributions are positive, the oracle's exact gap is

$$\frac{\pi_j f_j}{\pi_1 f_1 + \pi_2 f_2} > 0.$$

Proof. Substitute the oracle distribution in (1). In the singleton regime the result equals 1 precisely when the oracle assigns probability one to k , equivalently when the other contribution vanishes. $\square$

Corollary 2 (Strict-support conventions). *If A is a singleton, the infimum over strictly positive distributions is 1, but it is not attained. The family assigning probabilities $(1 - \eta, \eta)$ to the stopping and nonstopping items has expectation $1 + \eta$. If support is required only on positive-contribution items, the infimum is attained exactly when the nonstopping contribution is zero.*

Proof. Equation (1) is strictly decreasing in the nonstopping probability. Strict positivity forces that probability to be positive but permits it to approach zero. Under contribution support it may equal zero exactly when the nonstopping contribution is zero. $\square$

5 A certified counterexample

Take $N = 2$, $\varepsilon = 1/3$, $\delta = 1/20$, and

i	π_i	f_i	$\pi_i f_i$
1	3/4	1/3	1/4
2	1/4	1	1/4

Here $A = \{1\}$. The literal-simplex optimum audits item 1 with probability one and stops with certainty. The oracle is uniform because the contributions are equal, while prop-M samples item 1 with probability 3/4. Equation (1) gives

$$\mathbb{E}[\tau(q^{\text{opt}})] = 1 < \mathbb{E}[\tau(q^{\text{prop-M}})] = \frac{5}{4} < \mathbb{E}[\tau(q^{\text{oracle}})] = \frac{3}{2}.$$

The comparison $5/4 < 3/2$ uses two fully supported policies and therefore does not depend on the action-space ambiguity. Under strict full support, $q_\eta = (1 - \eta, \eta)$ further improves prop-M whenever $\eta < 1/4$, and approaches the infimum 1.

6 An infinite rational family

Let $p \in \mathbb{Q} \cap (1/2, 1)$, and set

$$(\pi_1, \pi_2) = (p, 1 - p), \quad (f_1, f_2) = \left(\frac{1 - p}{p}, 1 \right).$$

Choose rational ε with $1 - p \leq \varepsilon < p$, and any rational $\delta \in (0, 1)$. Both contributions equal $1 - p$, so the oracle is uniform and $A = \{1\}$. The theorem gives

$$\mathbb{E}[\tau(q^{\text{opt}})] = 1, \quad \mathbb{E}[\tau(q^{\text{prop-M}})] = 2 - p, \quad \mathbb{E}[\tau(q^{\text{oracle}})] = \frac{3}{2}.$$

Thus the oracle gap from the true literal-simplex optimum is $1/2$ throughout the family, while its gap from prop-M is $p - 1/2$.

7 Bellman formulation and verification

For $N \geq 3$, a state must record the ordered history, remaining set, wealth function, ApproxKelly accumulators, and running logical/probabilistic intersection. If $\Phi(x, q, i)$ is the next state after action q draws item i , and D is the stopping region, the remaining-cost Bellman equation is

$$V(x) = \begin{cases} 0, & x \in D, \\ 1 + \inf_q \sum_{i \in R(x)} q(i) V(\Phi(x, q, i)), & x \notin D. \end{cases}$$

The dependence of Φ on q through importance weighting is essential. Proposition 2 optimizes a local variance term, not the displayed continuation value.

The accompanying repository [3] represents every parameter and path probability as a reduced rational number. It enumerates all terminal histories, verifies all three support conventions, and independently regenerates a JSON certificate. No simulation, floating-point comparison, or numerical confidence-set grid is used.

The result is deliberately pinned to the released first-round initialization and logical intersection. It does not solve $N \geq 3$, a betting CS without the logical intersection, or the imperfect-information minimax problem.

References

- [1] S. Shekhar, Z. Xu, Z. Lipton, P. Liang, and A. Ramdas, Risk-limiting financial audits via weighted sampling without replacement, in *Proceedings of the 39th Conference on Uncertainty in Artificial Intelligence*, PMLR 216 (2023), 1932–1941.
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- [2] S. Shekhar and Z. Xu, *WeightedWoRConfSeq*, commit a834e459 (2023).
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- [3] R. Sater, *Exact Two-Transaction Sampling for Risk-Limiting Financial Audits* (2026).
<https://github.com/RichieSater/rlfa-optimal-policy>.